# Interactive Control of Curvature Extrema and Inflections on B-Spline Curves

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# Outline

1. **Sliding mechanism**

   Interactive editing under a curvature-extrema bound

2. **Closed curves**

   Ellipse, oval, ovoid, and the general 4-extrema shape space

3. **Complex rational B-splines**

   Möbius transformations

4. **PH curves**

   Lie sphere transformations

# Numerators of $\kappa$ and $\kappa'$

For a planar B-spline curve $\mathbf{c}(t) = \sum_i \mathbf{P}_i\, N_i^d(t)$:

- **Inflections** — zeros of $\kappa$, equivalently of $f(t) = \mathbf{c}'(t) \times \mathbf{c}''(t)$ (degree $2d-3$).
- **Curvature extrema** — zeros of $\kappa'$, equivalently of $g(t)$ (degree $4d-6$):

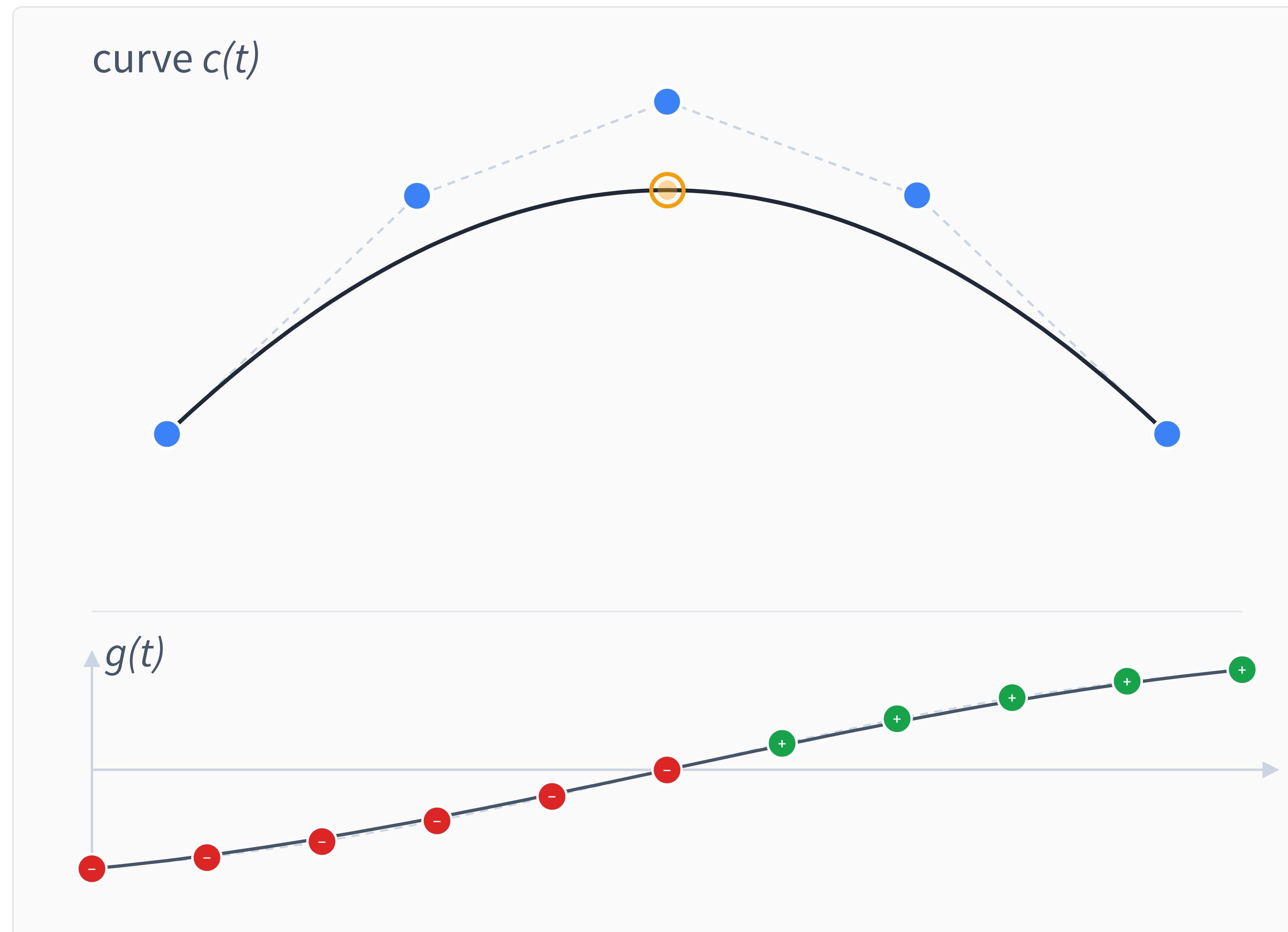
$$g(t) = \|\mathbf{c}'\|^2\,(\mathbf{c}' \times \mathbf{c}''') - 3\,(\mathbf{c}' \cdot \mathbf{c}'')\,(\mathbf{c}' \times \mathbf{c}'')$$

Both $f$ and $g$ are piecewise polynomial — computable exactly in Bernstein form via B-spline algebra.

# Without Sliding

Drag any control point. The bound on the number of curvature extrema is preserved — but the sign-change boundary in $g(t)$ is held rigid, so the extremum is **locked in** $t$.

This is how it worked before the sliding mechanism.



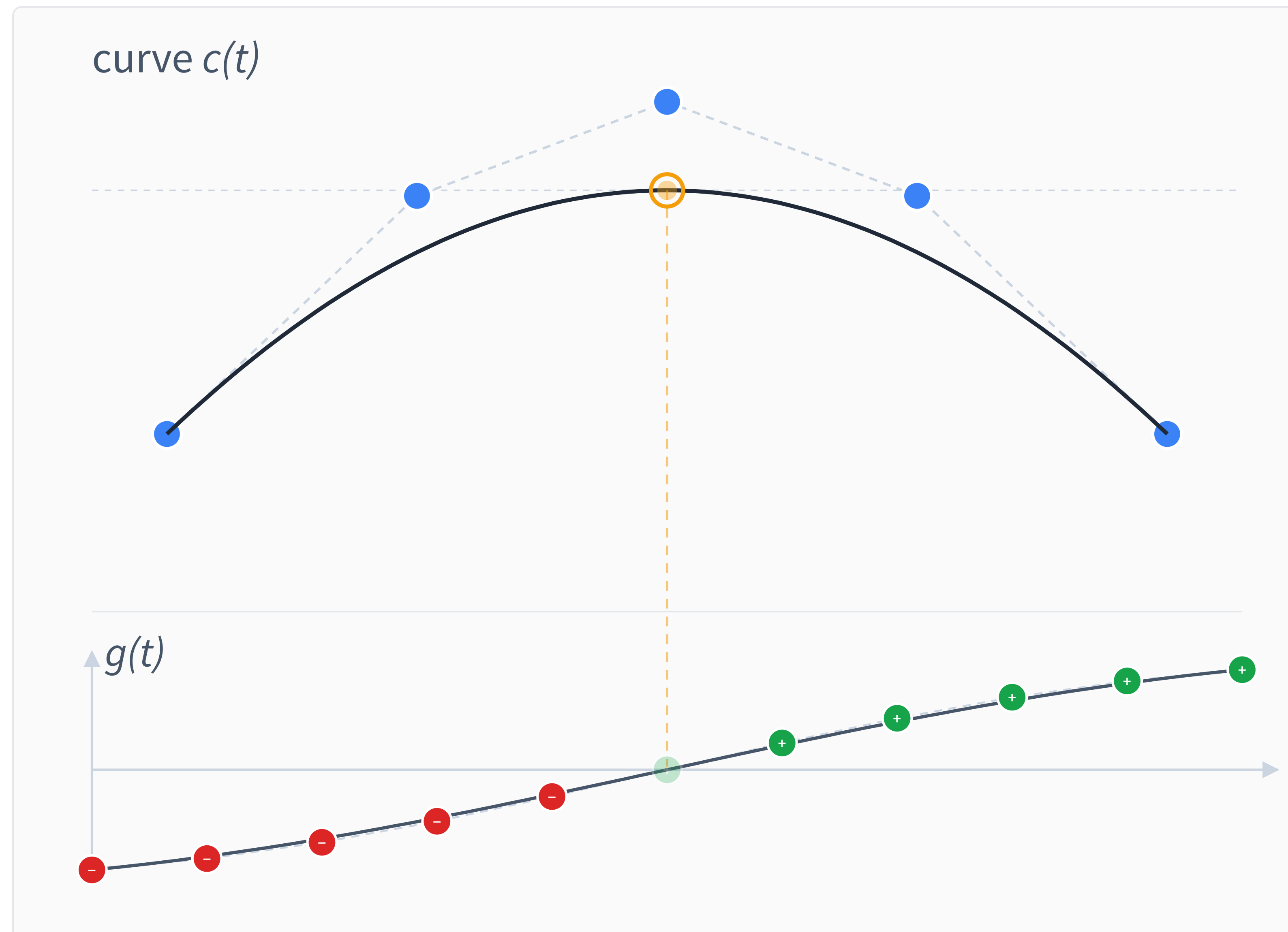

Bound extrema (S = 1)

# With Sliding

Drag the orange marker — the maximum of curvature.

Below: $g(t)$, the numerator of $\kappa'(t)$ — computable analytically. Its 11 Bernstein coefficients are shown (+ / −); faded dots are inactive constraints, free to slide.

One sign change, moving with the extremum.



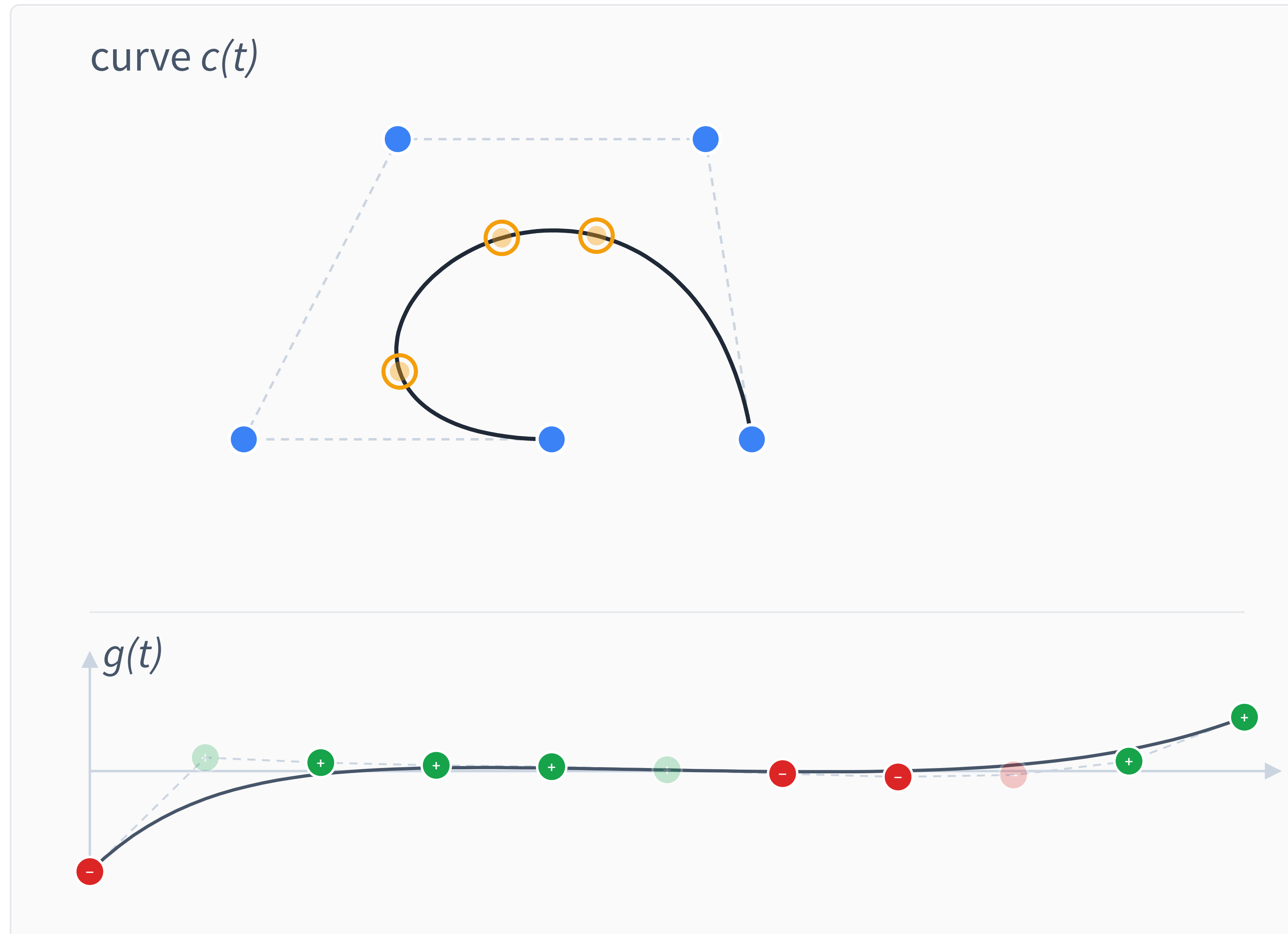
# Sliding and Annihilation

Three curvature extrema at the start. Drag the rightmost control point outward — two extrema slide toward each other and **annihilate**. The bound $S(\mathbf{b})$ drops.

The sliding mechanism makes $S(\mathbf{b})$ *monotone non-increasing*.

Bound extrema (S = 3)

Reset



# Theorems

**Theorem 1** (Schoenberg, variation diminishing). For a B-spline $g = \sum_i b_i N_i$,

$$S^-(g) \ \le \ S^-(\mathbf{b}).$$

Sign changes in the Bernstein coefficients $(b_i)$ bound the number of curvature extrema.

**Theorem 2** (contribution). Under constrained editing with the sliding mechanism, $S^-(\mathbf{b})$ is *monotone non-increasing*.

**Corollary.** The bound on the number of curvature extrema can only stay the same or decrease — across every single edit.

# Algorithm

When the user drags control point $\mathbf{P}_k$ toward target $\mathbf{T}_k$, solve:

$$\min_{\mathbf{P}^*} \sum_i w_i \left\|\mathbf{P}_i^* - \mathbf{T}_i\right\|^2 \quad \text{s.t.} \quad s_j \cdot g_j(\mathbf{P}^*) \geq 0,\ j \in \mathcal{A}.$$

The active set $\mathcal{A}$: positions with same-sign neighbors; one **anchor** per alternating sequence — the position with largest $|g|$. Solved by an interior-point method per drag step.

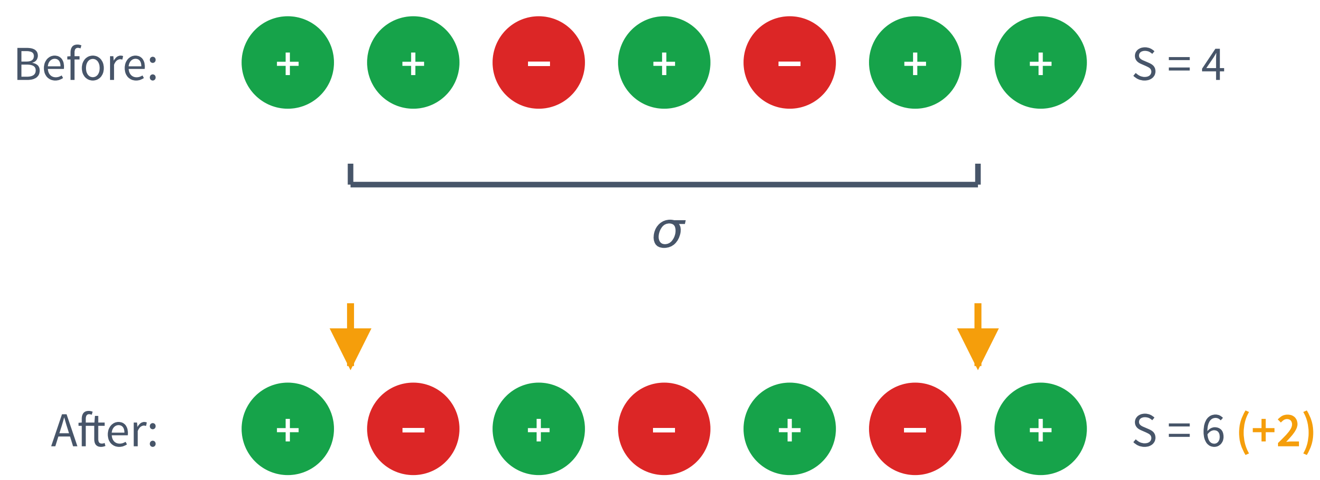
# Alternating sequence $\boldsymbol{\sigma}$

Inside an alternating sequence $\sigma$ of $\mathbf{b}$:

- **Interior**: $\sigma$ already has the *maximum possible* sign-change count (every adjacent pair is a sign change). Any flip can only reduce or preserve interior sign changes.
- **Junctions**: $\sigma$'s boundary pairs with the rest of the polygon are *not* sign changes. Each can *become* one if $\sigma$'s endpoint flips, at most $+1$ each.

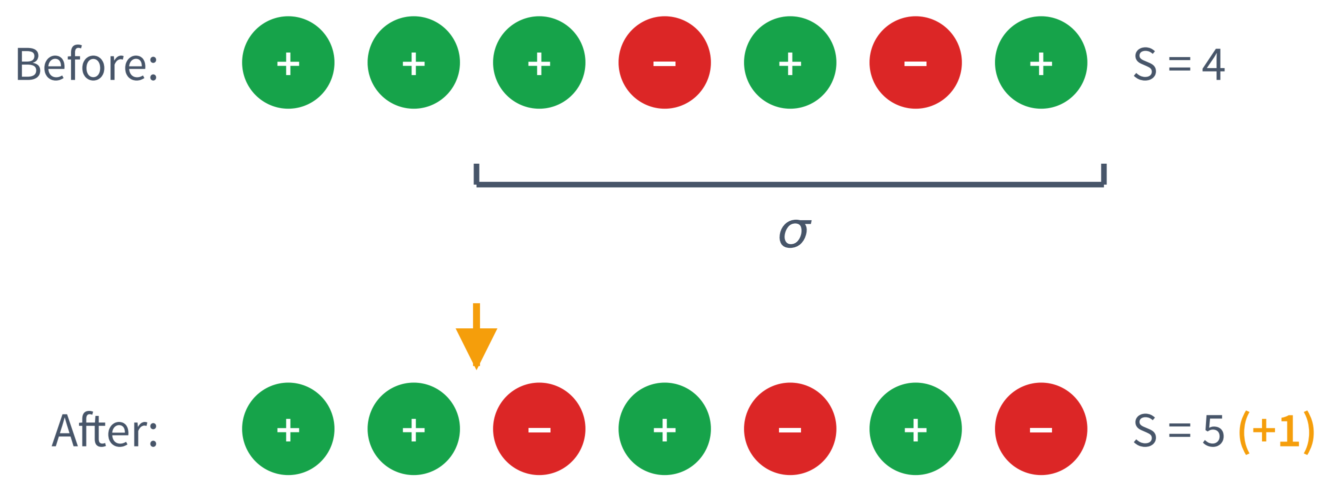
# All-flip

### Interior σ




*Both junctions become new sign changes.*

### Boundary σ

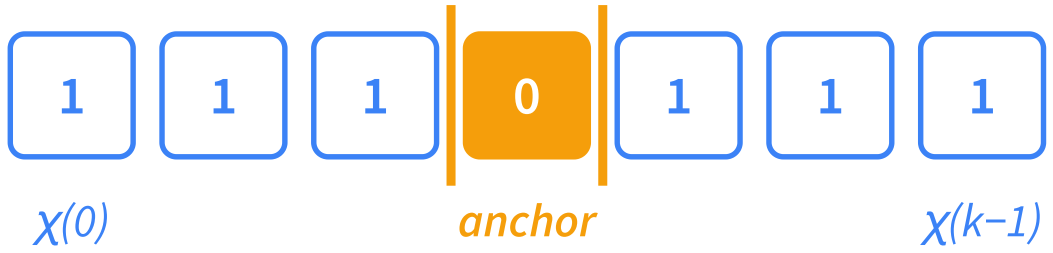



*Only one junction; the other end is the polygon boundary.*

*The only flip that grows $S^-$ is all-flip. Excluding any single position of **σ** from the flip — the **anchor** — blocks the all-flip. $S^-$ does not grow.*

# Anchor lemma

For any flip $G \subseteq \sigma$ that leaves at least one sign fixed (the **anchor**), $S^-(\mathrm{flip}_G\, \mathbf{b}) \le S^-(\mathbf{b})$.




*Proof.* Encode the flip as a boolean pattern $\chi$ on $\sigma$, 1 if sign is flipped, 0 otherwise.

Each *transition* of $\chi$ destroys one interior sign change; $T(\chi)$ counts the transitions.

$$S^-(\mathrm{flip}_G\, \mathbf{b}) - S^-(\mathbf{b}) \;\le\; \chi(0) + \chi(k-1) - T(\chi).$$

With one anchor the right-hand side is $\le 0$. ■

# Multiple sequences

For a polygon $\mathbf{b}$ with alternating sequences $\sigma_1, \ldots, \sigma_n$, **one anchor per sequence** gives safety: $S^-(\operatorname{flip}\mathbf{b}) \leq S^-(\mathbf{b})$.

*Proof.* Apply the anchor lemma to each $\sigma_i$ independently. A shared junction between adjacent $\sigma_i$ and $\sigma_{i+1}$ contributes via XOR:

$$\chi_i(\text{last}) \oplus \chi_{i+1}(0) \;\leq\; \chi_i(\text{last}) + \chi_{i+1}(0),$$

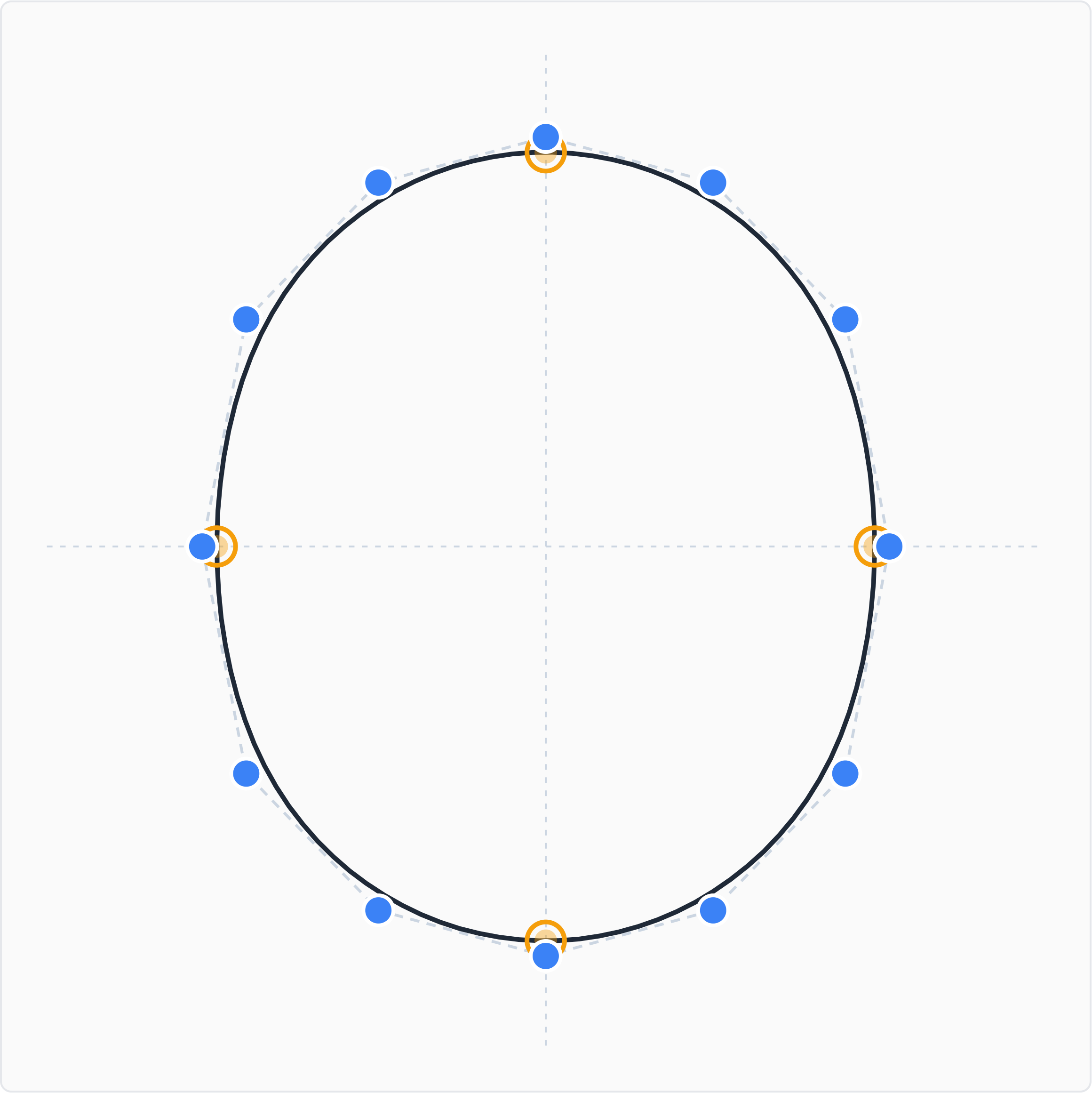
so the per-sequence bounds sum without double-counting. Each is $\leq 0$ by the anchor lemma, so the total is $\leq 0$. ■

# Oval

A closed curve with

- **2** axes of symmetry
- **4** curvature extrema
- **0** inflections

Drag any control point.

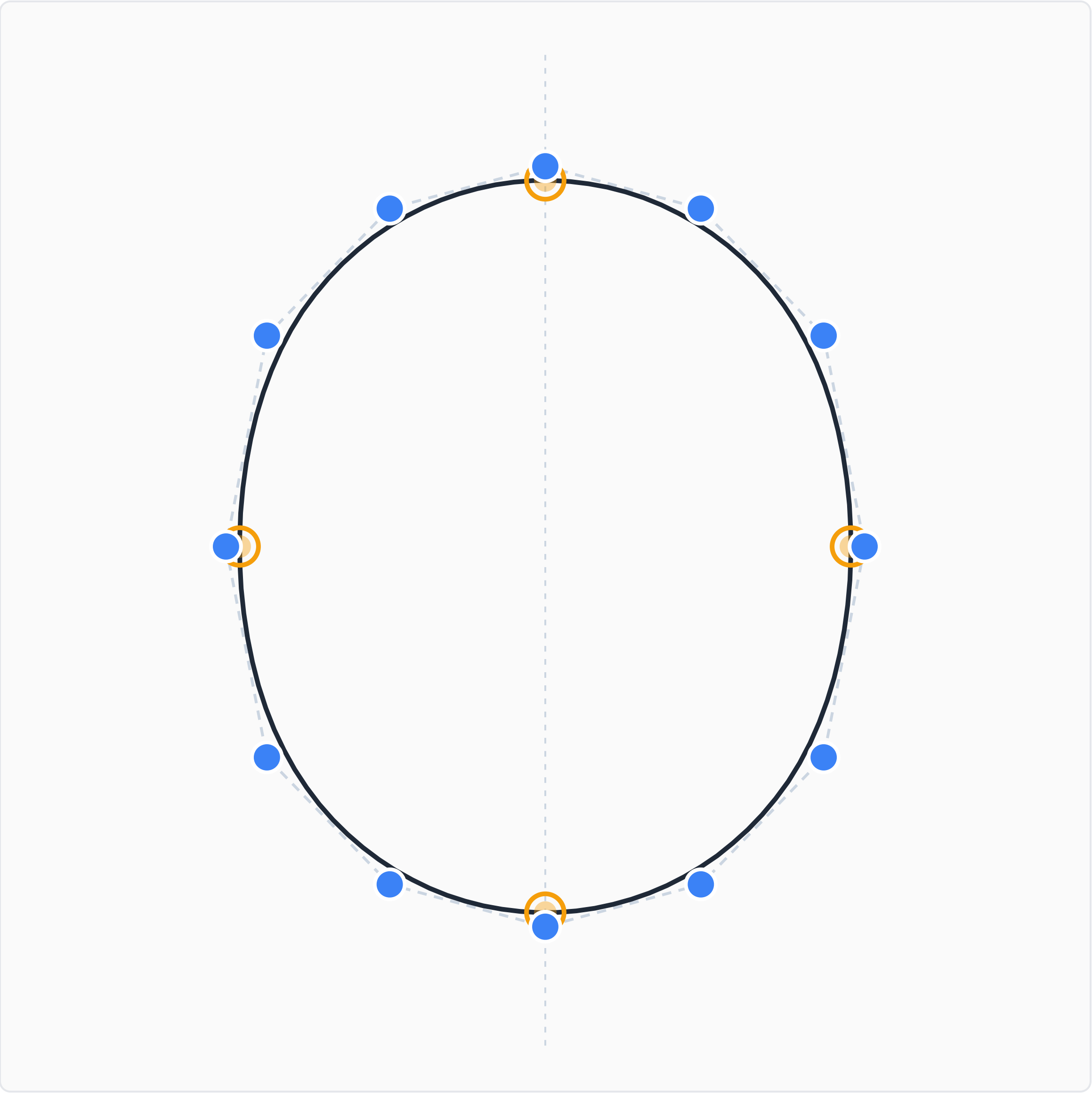


# Single axis of symmetry

A closed curve with

- **1** axis of symmetry
- **4** curvature extrema
- inflections *free*

Drag any control point.



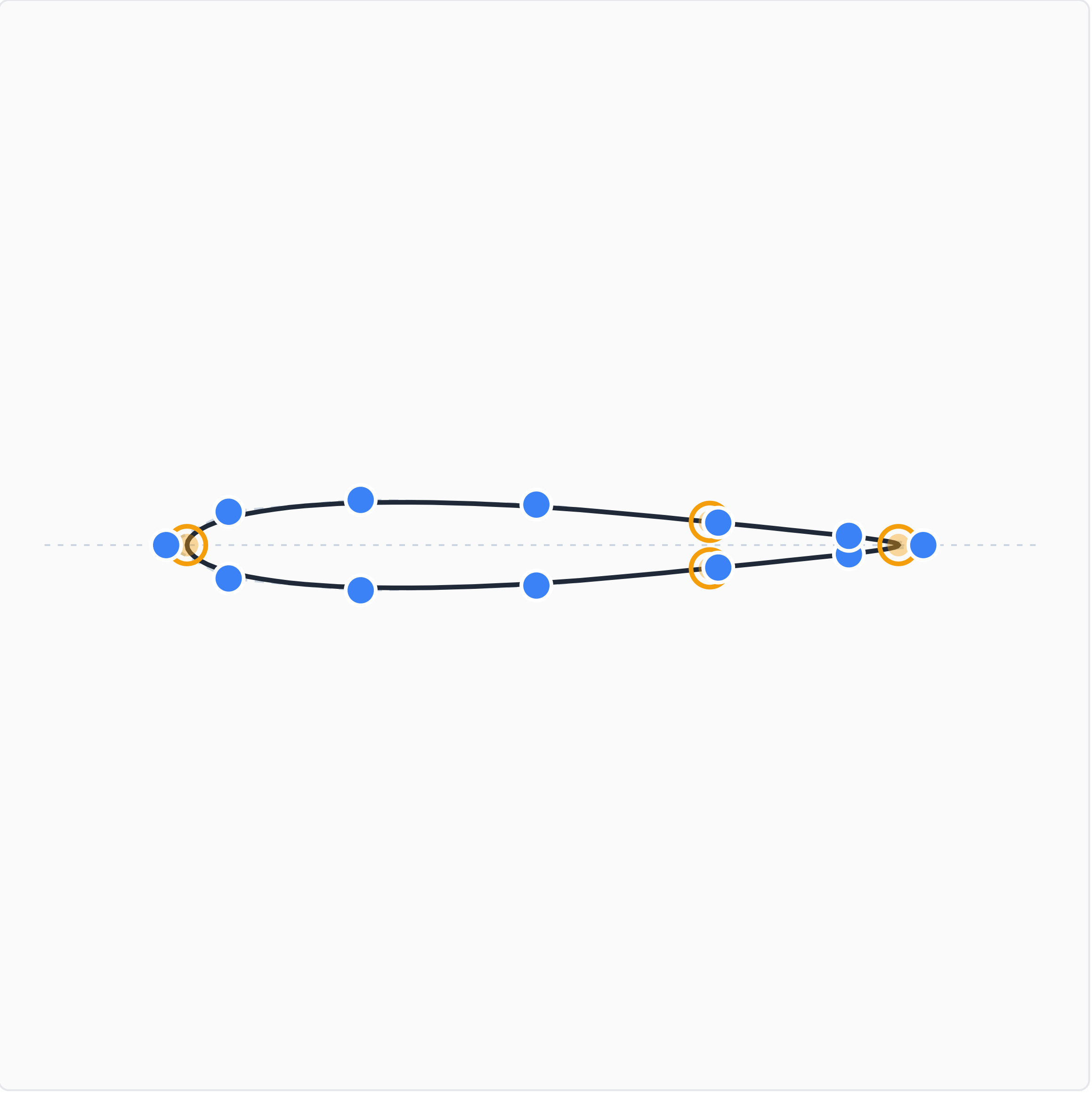
# Airfoil

A closed curve with

- **4** curvature extrema
- inflections *free*
- x-axis symmetry — toggleable

Symmetric

Drag any control point.

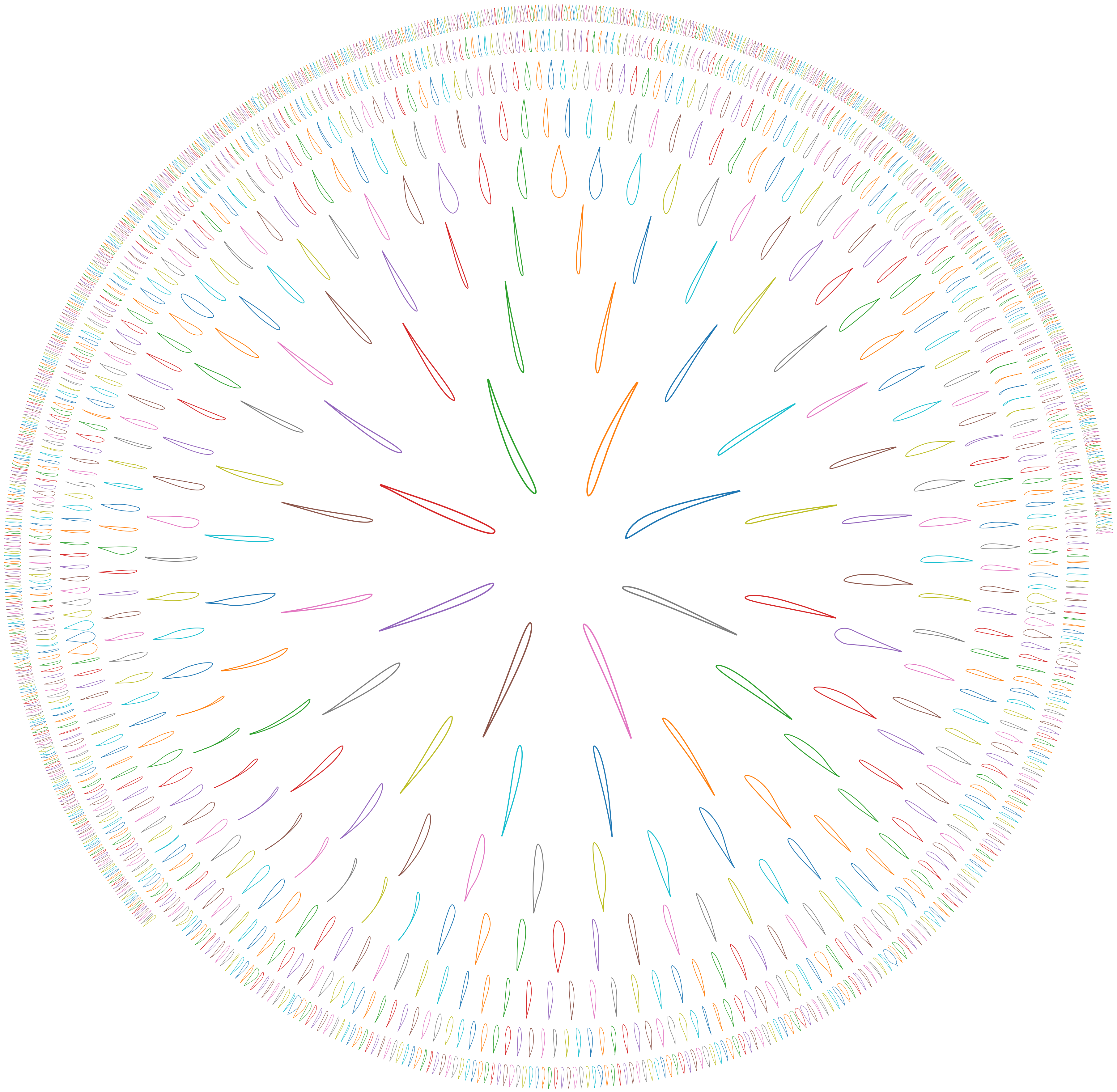


# The UIUC airfoil dataset

**1644** airfoils, same machinery.

- **98.8%** admit a 4-curvature-extrema fit
- median error **0.13% chord** (p90 = 0.38%)
- **0** errors, **0** fits worsened by the constraint

*Workflow: fair → fit with an interior-point optimizer → a primal–dual solve tightens the fit.*

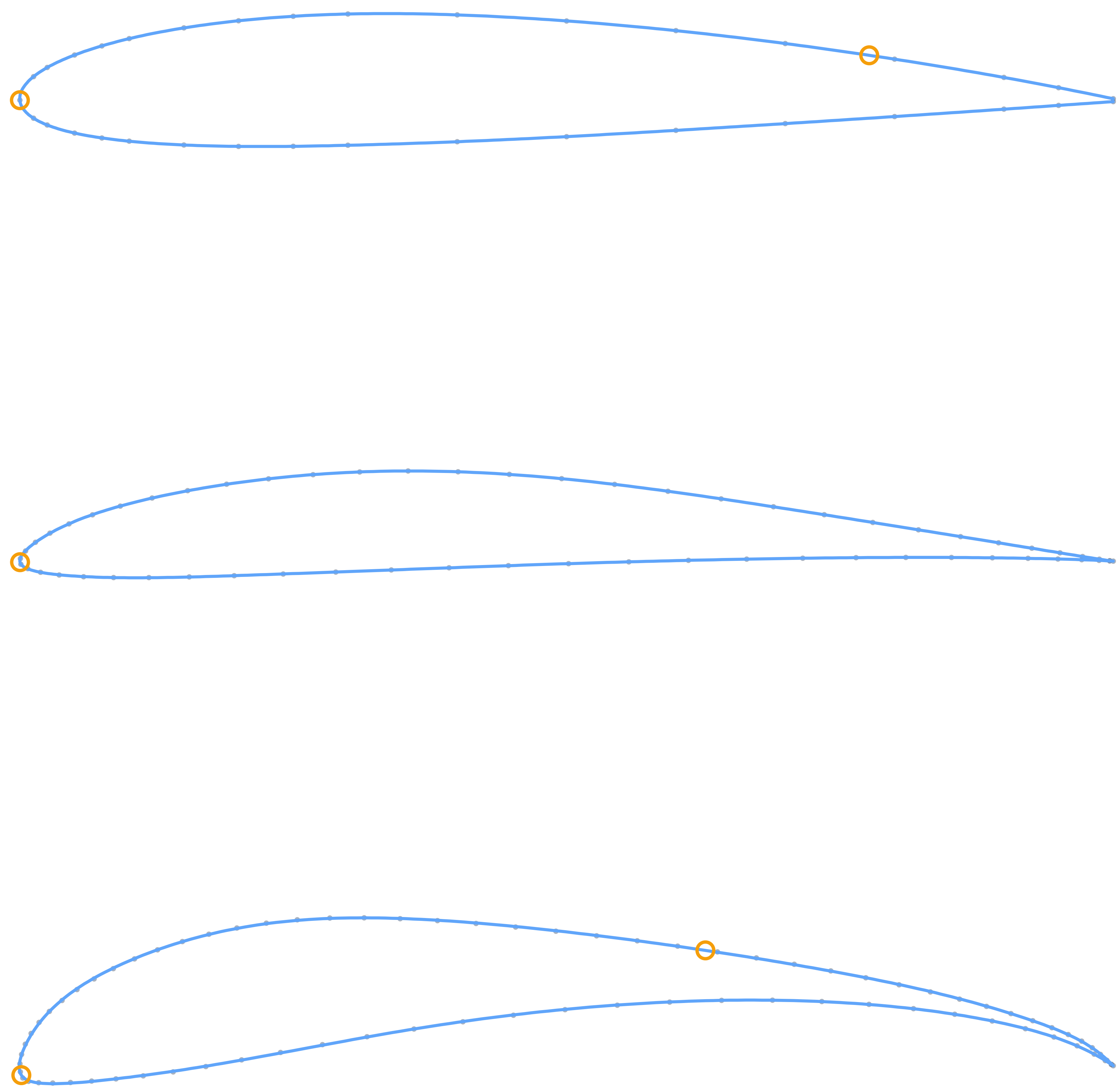


# How precise can we get?

Open curve, trailing edge pinned — push the bounded fit to its limit.

- degree-5 B-spline · *Fair* ↔ *Fit* · primal–dual finish
- 1–2 curvature extrema, **< 0.1% chord**

*An AI-assisted algorithm-design experiment (Claude Code).*



NACA 2412
2 curvature extrema
error 0.008% chord

Eppler E387
1 curvature extremum
error 0.019% chord

Selig S1223 (high-lift)
2 curvature extrema
error 0.071% chord

*Open degree-5 B-spline, trailing edge pinned · curvature-extrema bound held throughout · data dots = raw UIUC coordinates*

# Complex Rational B-spline

Move the slider to apply Möbius transformation.

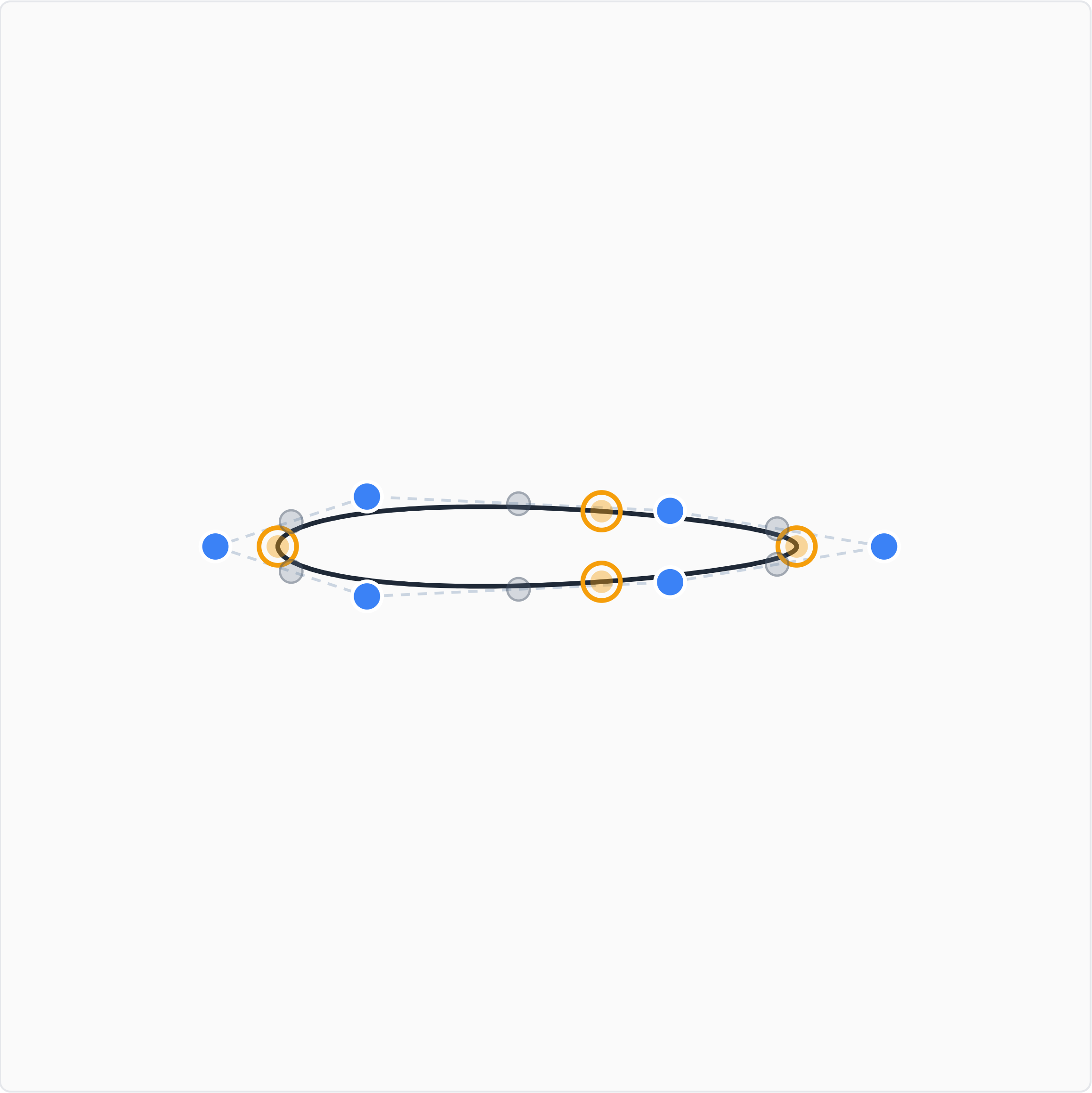
Bound extrema (S = 4)

θ 0°

inv x 0.00

inv y 0.00

Apply Reset



# PH curves & Lie sphere transformations

A **complex-rational Pythagorean-hodograph** curve $z = A/B$ is built so its hodograph generator is a perfect square — $A'B - AB' = S^2$ — making the speed $|z'| = |S|^2/|B|^2$ rational. So it is PH *by construction*: rational offsets, rational arc length.

- The same $\kappa'$ sign-change bound applies on the PH curve — edit the generator, the curvature-extrema count is monotone non-increasing.
- Revolve it into a **canal surface**: its **ridges** (curvature extrema of the principal family) are exactly the curve's extrema, swept into rings.
- **Lie sphere transformations** act linearly on the oriented-contact lift and *preserve ridges* — a further bound-preserving editing group, now on the surface.

Open the Lie Sphere Workbench →

# Conclusion

**Sign changes of $\kappa'$ and $\kappa$ are bounded — monotone non-increasing under editing.**

1. **Sliding mechanism + knot insertion**
2. **Closed curves** — the 4-extrema shape space: ellipse → oval → ovoid → general
3. **Complex rational B-splines** — Möbius transformations
4. **PH curves** — Lie sphere transformations

# References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# Thank You

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Questions?

# Backup

The exact sparse Jacobian of g · the PH curves we edit (A, B, S and S, D)

# Differentiating with Respect to a Control Point

A B-spline curve is $\mathbf{c}(t) = \sum_j \mathbf{P}_j \, N_j(t)$.

Differentiate with respect to control point $\mathbf{P}_i$:

$$\frac{\partial \mathbf{c}(t)}{\partial \mathbf{P}_i} = N_i(t)$$

This is itself a B-spline — with the same knot vector and degree, but with control points:

```
0, 0, ..., 0, 1, 0, ..., 0
```

A single nonzero control point at position $i$. Almost entirely zero.

# Derivatives Are B-Splines Too

Differentiate $N_i(t)$ with respect to $t$:

- $N_i'(t)$ is a B-spline of degree $d{-}1$
- $N_i''(t)$ is a B-spline of degree $d{-}2$
- $N_i'''(t)$ is a B-spline of degree $d{-}3$

All have the **same local support** — nonzero on only $d{+}1$ consecutive spans out of $n$.

And they depend only on the knot vector and degree, not on the control point positions. Computed **once**, cached for all optimization iterations.

# No Need to Multiply Zeros

To compute $\partial g / \partial x_i$, apply the product rule to $g$. Each term is a product of:

- Intermediate products already computed for $g$ — defined on all $n$ spans, **shared** across all columns
- A cached basis derivative $N_i'$ or $N_i''$ or $N_i'''$ — nonzero on only $d{+}1$ spans

Multiply only on the $d{+}1$ nonzero spans. Skip the rest.

For $n = 100, d = 3$: each Jacobian column requires work on **4 spans** instead of 97.

Result: exact Jacobian, **3–5× faster** than automatic differentiation — which does not know what is zero.

# The PH curve we edit: A, B, S

**Complex-rational Pythagorean-hodograph** splines are defined by $z(t) = A(t)/B(t)$, with $A, B$ complex B-splines and a generator $S$ enforcing the *PH condition*:

$$A'B - AB' = S^2$$

The hodograph $z' = S^2/B^2$ is a perfect square, so the square root inside the norm collapses:

$$|z'| = \sqrt{z'\,\overline{z'}} = \sqrt{\frac{S^2}{B^2} \cdot \frac{\overline{S}^2}{\overline{B}^2}} = \sqrt{\left(\frac{S\overline{S}}{B\overline{B}}\right)^2} = \frac{S\overline{S}}{B\overline{B}}$$

The speed is **rational** — hence rational offsets.

# Editing keeps it PH: a constrained solve

Both sides of the PH condition are degree $2p-2$, so they live in the same **Bernstein basis**. Equality means matching coefficients — equivalently, the residual must vanish:

$$R = A'B - AB' - S^2 = 0 \quad \text{(every Bernstein coefficient)}$$

So editing is a **constrained optimization**, not a formula. The control points $A, B, S$ are *all* free (one weight $B_0$ pinned as a gauge), and an interior-point solver moves them to:

- **minimize** the distance from the dragged control point to the cursor,
- **subject to** $R = 0$ (stays a PH curve) and the sign constraints $s_j\, g_j \geq 0$ (keeps the curvature-extrema count).

Letting $S$ float alongside $A, B$ is what lets the curve bend, rotate and scale freely while never leaving the PH family.

# Or: PH by construction — the (S, D) form

A second route drops the constraint entirely. Write $z = F/D$ and choose the **generator** $S$ and denominator $D$ as the free variables. The hodograph numerator is fixed to a square by *fiat*:

$$H = S^2 \quad \Longrightarrow \quad F'D - FD' = S^2 \ (\mathrm{Wronskian})$$

This is **linear in** $F$, so the numerator is simply *recovered* — by integration when $D$ is constant ( $F = \int (S/D)^2$), or a small per-span linear solve otherwise.